Halo-model derivation of $\langle \Delta\Sigma^{\rm prj}(R \mid \lambda^{\rm ob}, z^{\rm ob})\rangle$: can we avoid a second integral?

Johnny H. Esteves

May 6, 2026

Abstract

We revisit the two-halo projected surface density $\langle\Sigma^{\rm prj}\rangle$ of Costanzi et al. (2026) and derive it as the cylindrical specialisation of the Cooray–Sheth halo-halo two-halo term (Cooray & Sheth (2002), Eqs. 86–87). We then ask a practical question: to get $\langle\Delta\Sigma^{\rm prj}(R)\rangle$, do we need a new integral, or can we reuse the $\langle\Sigma^{\rm prj}(R)\rangle$ machinery? The answer is that the radial-average and subtraction $\bar{\Sigma}(< R) - \Sigma(R)$ is a linear functional in R only, so it commutes with the (θ, z, M) integrals: the correct recipe is to swap $\Sigma_{\rm mis} \to \Delta\Sigma_{\rm mis}$ inside the Eq. 13 kernel and leave everything else alone. We also argue that, because $\Delta\Sigma_{\rm mis}(R' \mid R_{\rm mis}) \to 0$ for $R_{\rm mis} \gg R'$ (the surface-density contrast of a distant off-centre NFW neighbour vanishes), the θ-upper limit of the integral can be reduced from the $R_{\rm max} = 30\,h^{-1}\,$Mpc legacy value to a physics-driven $\theta_{\rm max} \sim$ a few $\times\, R'_{\rm max}/\chi(z_{\rm cls})$, set by the support of $\Delta\Sigma_{\rm mis}$ rather than by $\xi_{\rm NL}$ decay or by the `twoD_prj_NFW` truncation.

1 Setup: Cooray–Sheth two-halo term in cylindrical coordinates

Geometry and notation. Fix the cluster at the origin, at comoving distance $\chi(z_{\rm cls})$; its mass is $M_{\rm cls}$ and its halo bias $b_{\rm cls} \equiv b(M_{\rm cls}, z_{\rm cls})$. A neighbouring halo (the two-halo contributor, "2h") sits at sky angle θ from the cluster and redshift z, with mass M and halo bias $b(M, z)$. We adopt cylindrical coordinates $(R,\, \chi_\parallel)$ around the cluster's line of sight, with

$$R \equiv \theta\,\chi(z) \;\approx\; \theta\,\chi(z_{\rm cls}), \qquad \chi_\parallel \equiv \chi(z) - \chi(z_{\rm cls}), \qquad r \equiv \sqrt{R^2 + \chi_\parallel^2}, \tag{1}$$

where the small-angle approximation $R \simeq \theta\,\chi(z_{\rm cls})$ holds across the photo-z window ($|\chi_\parallel| \ll \chi(z_{\rm cls})$) and r is the 3-D comoving cluster–neighbour separation. The exact form used numerically (`sigma_prj.py`, §3 of the refactor note) is the chord

$$r^2 \;=\; \chi(z)^2 + \chi(z_{\rm cls})^2 - 2\,\chi(z)\,\chi(z_{\rm cls})\cos\theta, \tag{2}$$

which reduces to (1) at $\theta \ll 1$.

Starting point: Cooray–Sheth Eq. 86. The 3-D matter two-halo correlator is (Cooray & Sheth, 2002, Eq. 86)

$$\xi_{2h}(\mathbf{x} - \mathbf{x}') = \int dM_{\rm cls}\,\frac{M_{\rm cls} n(M_{\rm cls})}{\bar\rho} \int dM\,\frac{M\,n(M)}{\bar\rho} \times \int d^3x_c\, u(\mathbf{x} - \mathbf{x}_c \mid M_{\rm cls}) \int d^3x\, u(\mathbf{x}' - \mathbf{x} \mid M)\,\xi_{hh}(\mathbf{x}_c - \mathbf{x} \mid M_{\rm cls}, M), \tag{3}$$

with $\mathbf{x}_c$ the cluster centre, $\mathbf{x}$ the 2h halo centre, and $u(\cdot \mid M)$ the unit-normalised density profile. For our observable we fix the cluster centre at the origin, $\mathbf{x}_c \equiv \mathbf{0}$, and drop its own profile (the 1-halo term is treated separately). What remains is the surface density around the cluster contributed by projected neighbouring halos:

$$\Sigma^{\rm prj}(R \mid M_{\rm cls}, z_{\rm cls}) = \int d^2R_\perp \int d\chi_\parallel \int dM\, n(M,z)\, \Sigma_{\rm NFW}\big(|\mathbf{R} - \mathbf{R}_\perp| \,\big|\, M, z\big)\, \xi_{hh}\big(r \,\big|\, M_{\rm cls}, M\big), \tag{4}$$

where $\mathbf{R}_\perp$ is the transverse offset of the neighbour from the cluster line of sight ($|\mathbf{R}_\perp| = R$), $\chi_\parallel$ is its LoS offset, and $\Sigma_{\rm NFW}(R' \mid M, z)$ is the centred NFW surface density of the neighbour evaluated at 2-D radius R' from its own centre.

Change of coordinates to (θ, z). Swap the neighbour's $(\mathbf{R}_\perp, \chi_\parallel)$ for the observable pair (θ, z). The volume element becomes

$$d^2R_\perp\, d\chi_\parallel \;=\; 2\pi \sin\theta\, d\theta \cdot \frac{dV}{dz\, d\Omega}(z)\, dz. \tag{5}$$

Azimuth-averaging $\Sigma_{\rm NFW}(|\mathbf{R} - \mathbf{R}_\perp|)$ around the neighbour's off-centre offset $R_{\rm mis} \equiv R$ produces the *miscentered* NFW surface density,

$$\Sigma_{\rm mis}(R' \mid M, z, R_{\rm mis}) = \int_0^{2\pi} d\varphi\, \Sigma_{\rm NFW}\Big(\sqrt{R'^{\,2} + R_{\rm mis}^2 - 2R'R_{\rm mis}\cos\varphi} \,\Big|\, M, z\Big), \tag{6}$$

where R' is the radius at which the cluster observer measures $\Sigma^{\rm prj}$. For the cluster–neighbour geometry $R_{\rm mis} = \theta\,\chi(z_{\rm cls})$ (equation 1). Equation (4) becomes

$$\Sigma^{\rm prj}(R' \mid M_{\rm cls}, z_{\rm cls}) = \int d\theta \sin\theta \int dz\, \frac{dV}{dz\, d\Omega}(z) \int dM\, n(M,z) \\ \times\; \xi_{hh}\big(r \mid M_{\rm cls}, M\big)\, \Sigma_{\rm mis}\big(R' \mid M, z, R_{\rm mis} = \theta\,\chi(z_{\rm cls})\big), \tag{7}$$

with r given by (2).

Cooray–Sheth Eq. 87 step. Deterministic linear halo bias replaces the halo-halo correlator by matter clustering times two bias factors, $\xi_{hh}(r \mid M_{\rm cls}, M) \approx b_{\rm cls}\, b(M,z)\, \xi_{\rm lin}(r)$. Factoring $b_{\rm cls}$ out:

$$\Sigma^{\rm prj}(R' \mid M_{\rm cls}, z_{\rm cls}) \approx b_{\rm cls} \int d\theta \sin\theta \int dz\, \frac{dV}{dz\, d\Omega} \int dM\, n(M,z)\, b(M,z) \\ \times\; \xi_{\rm lin}(r)\, \Sigma_{\rm mis}\big(R' \mid M, z, \theta\,\chi(z_{\rm cls})\big). \tag{8}$$

From mass-selected to richness-selected. Costanzi et al. (2026) upgrade (8) in three steps to describe the observable around an optically selected cluster of observed richness $\lambda^{\rm ob}$: (i) $\xi_{\rm lin} \to \xi_{\rm NL}$ (halofit), because the 1h–2h transition at $\sim R_{\rm excl}$ sits in the non-linear regime; (ii) $\xi_{\rm NL} \to 0$ for $\theta < \theta_{\rm excl}(z)$ (LoS-slab exclusion); (iii) the cluster's bias $b_{\rm cls}$ is replaced by the scale-dependent *selection-affected* bias $b_{\rm sel}(\theta; \lambda^{\rm ob}, z^{\rm ob})$, which absorbs red-galaxy selection effects. Adding the uncorrelated cosmological mean as a $+1$ inside the bracket yields the master equation (Costanzi et al. 2026, Eq. 13):

$$\langle \Sigma^{\rm prj}(R' \mid \lambda^{\rm ob}, z^{\rm ob}) \rangle = \int d\theta \sin\theta \int dz\, \frac{dV}{dz\, d\Omega} \int dM\, n(M,z) \\ \times\; \big[\, 1 + b(M,z)\, b_{\rm sel}(\theta; \lambda^{\rm ob}, z^{\rm ob})\, \xi_{\rm NL}(r, z^{\rm ob}) \,\big]\, \Sigma_{\rm mis}\big(R' \mid M, z, R_{\rm mis} = \theta\,\chi(z_{\rm cls})\big). \tag{9}$$

The 1 piece integrates to a spatially near-uniform $\Sigma_{\rm rnd}(R')$ (the cosmological mean projected surface density in the photo-z window); the $b\, b_{\rm sel}\, \xi_{\rm NL}$ piece is the correlation-excess two-halo contribution $\Sigma_{\rm cl+LSS}(R')$.

2 The target observable: $\langle\Delta\Sigma^{\rm prj}(R')\rangle$

The lensing observable is the *excess* surface density,

$$\langle\Delta\Sigma^{\rm prj}(R')\rangle \equiv \bar{\Sigma}^{\rm prj}(<R') - \langle\Sigma^{\rm prj}(R')\rangle, \qquad \bar{\Sigma}^{\rm prj}(<R') \equiv \frac{2}{R'^2}\int_0^{R'} s\,\langle\Sigma^{\rm prj}(s)\rangle\,ds. \tag{10}$$

Linearity in R'. Both $\bar{\Sigma}(<R')$ and the subtraction act only on the radial argument R'. The outer integrals in (9) (over θ, z, M) are R'-independent, so they commute with $\Delta[\,\cdot\,]$:

$$\langle\Delta\Sigma^{\rm prj}(R' \mid \lambda^{\rm ob}, z^{\rm ob})\rangle = \int d\theta \sin\theta \int dz\,\frac{dV}{dz\,d\Omega}\int dM\,n(M,z) \\ \times\ \big[\,1 + b(M,z)\,b_{\rm sel}(\theta;\lambda^{\rm ob},z^{\rm ob})\,\xi_{\rm NL}(r,z^{\rm ob})\,\big]\,\Delta\Sigma_{\rm mis}\big(R' \mid M,z,R_{\rm mis}=\theta\,\chi(z_{\rm cls})\big), \tag{11}$$

where the only change from (9) is

$$\Sigma_{\rm mis}\big(R' \mid M,z,R_{\rm mis}\big) \;\longrightarrow\; \Delta\Sigma_{\rm mis}\big(R' \mid M,z,R_{\rm mis}\big) \equiv \bar{\Sigma}_{\rm mis}(<R' \mid M,z,R_{\rm mis}) - \Sigma_{\rm mis}(R' \mid M,z,R_{\rm mis}). \tag{12}$$

The machinery that computes the (θ, z, M) integral (the split-at-breakpoints θ-grid with per-R' nodes, the photo-z z-slab, the exact r form, the exclusion cut) is preserved — $\Delta\Sigma_{\rm mis}$ just replaces $\Sigma_{\rm mis}$ as a lookup against the offset-NFW tables.

Does the $\Sigma_{\rm rnd}$ piece drop out? In the idealised sky-limit $\theta_{\max}\to\infty$ with infinitely narrow 1-halo profiles, the 1 piece gives a spatially uniform $\Sigma_{\rm rnd}$, for which $\bar{\Sigma}(<R') \equiv \Sigma_{\rm rnd}$ and thus $\Delta\Sigma_{\rm rnd}(R') = 0$. This is the classical Sheldon et al. (2009)–Zu et al. (2014)–Melchior et al. (2017) statement that the mean photo-z-window surface density cancels from the measured excess. Numerically the package truncates θ at $\theta_{\max} = R_{\max}/\chi(z_{\rm cls})$ with $R_{\max} = 30\,h^{-1}\,$Mpc (`config.R_MAX_CMPCH`), so $\Delta\Sigma_{\rm rnd}$ is not exactly zero but a boundary term that scales with $R_{\max}$. We therefore recommend, by analogy with $\langle\Sigma^{\rm prj}\rangle$, returning only the cl+LSS piece by default:

$$\langle\Delta\Sigma^{\rm prj}_{\rm cl+LSS}(R')\rangle = \int d\theta \sin\theta \int dz\,\frac{dV}{dz\,d\Omega}\int dM\,n(M,z) \\ \times\ b(M,z)\,b_{\rm sel}(\theta)\,\xi_{\rm NL}(r,z^{\rm ob})\,\Delta\Sigma_{\rm mis}\big(R' \mid M,z,\theta\,\chi(z_{\rm cls})\big). \tag{13}$$

3 Choosing $\theta_{\max}$ for $\langle\Delta\Sigma^{\rm prj}\rangle$

The θ-upper limit of (9)/(11) is an artefact of numerical truncation; the physics runs to $\theta\to\pi$. For $\langle\Sigma^{\rm prj}\rangle$ the legacy choice is $R_{\max} = 30\,h^{-1}\,$Mpc, so that $\theta_{\max} = R_{\max}/\chi(z_{\rm cls})$, matching the hard truncation in Matteo's `twoD_prj_NFW` notebook (`config.R_MAX_CMPCH`, §5.1 of the refactor note). At that scale the `cl+LSS` piece is converged to $<1\%$ because $\xi_{\rm NL}(r)$ has decayed, but the `rnd` piece grows by $\sim 15\%$ going $30\to 60\,h^{-1}\,$Mpc: the flat large-θ plateau of the `rnd` integrand never dies (see Esteves, 2026, §8). That is why the package returns `cl+LSS` by default for $\langle\Sigma^{\rm prj}\rangle$.

What changes for $\langle\Delta\Sigma^{\rm prj}\rangle$. Two facts together imply that the $\theta_{\max}$ requirement for $\langle\Delta\Sigma^{\rm prj}\rangle$ is strictly *weaker* than for $\langle\Sigma^{\rm prj}\rangle$:

1. **The `rnd` piece is dropped by design.** Following (13), $\langle\Delta\Sigma^{\rm prj}\rangle$ in this package is the `cl+LSS`-only integral — the long θ-plateau of the `rnd` piece simply does not appear in the observable.

2. **The `cl+LSS` θ-integrand of $\langle\Delta\Sigma^{\rm prj}\rangle$ is itself bounded in θ.** $\Delta\Sigma_{\rm mis}(R' \mid R_{\rm mis})$ has compact support around $R_{\rm mis} \sim R'$ and decays in both limits ($R_{\rm mis} \ll R'$ gives $\Delta\Sigma_{\rm mis} \to 2\pi\Delta\Sigma_{\rm NFW}(R')$, falling in R'; $R_{\rm mis} \gg R'$ gives $\Sigma_{\rm mis} \to 2\pi\Sigma_{\rm NFW}(R_{\rm mis})$, R'-independent, so $\Delta\Sigma_{\rm mis} = \bar{\Sigma}_{\rm mis} - \Sigma_{\rm mis} \to 0$). In contrast, the $\langle\Sigma^{\rm prj}\rangle$ `cl+LSS` integrand inherits the slow plateau of $\Sigma_{\rm mis}(R' \mid R_{\rm mis})$ at large $R_{\rm mis}$, which in numerical plots shows up as a near-constant $N_{\rm cl+LSS}(\theta)$ beyond θ_λ.

In other words, the two mechanisms that forced $\theta_{\rm max} = 30\,h^{-1}\,{\rm Mpc}/\chi(z_{\rm cls})$ for $\langle\Sigma^{\rm prj}\rangle$ — the `rnd` plateau *and* the flat $\Sigma_{\rm mis}$-kernel at large $R_{\rm mis}$ — both go away when we switch to $\Delta\Sigma^{\rm prj}_{\rm cl+LSS}$. The remaining constraint on $\theta_{\rm max}$ is only that we resolve the support of $\Delta\Sigma_{\rm mis}(R' \mid \theta\,\chi(z_{\rm cls}))$, which is set by the largest measurement radius $R'_{\rm max}$.

Recipe. Take $\theta_{\rm max}$ as the first of $\{\theta_{R'_{\rm max}}$ times a safety factor, θ_λ times a safety factor$\}$ that is large enough to include the tail of the ring peak at $R_{\rm mis} = R'_{\rm max}$. Empirically,

$$\boxed{\theta^{\Delta\Sigma}_{\rm max} = \frac{C\,R'_{\rm max}}{\chi(z_{\rm cls})}, \qquad C \simeq 3,\ R'_{\rm max} \equiv \max_i R'_i.} \tag{14}$$

For a DES-Y3-style setup with $R'_{\rm max} = 10\,h^{-1}\,{\rm Mpc}$ and $z_{\rm cls} = 0.5$, $\theta^{\Delta\Sigma}_{\rm max} \approx 30/\chi(z_{\rm cls})$, so numerically it happens to coincide with the $\langle\Sigma^{\rm prj}\rangle$ default; for $R'_{\rm max} = 3\,h^{-1}\,{\rm Mpc}$ it drops to $\sim 9/\chi(z_{\rm cls})$, a $\sim 3\times$ cheaper θ-integral. Do *not* push C below ~ 2: the ring peak of $\Delta\Sigma_{\rm mis}$ at $R_{\rm mis} = R'_{\rm max}$ extends modestly beyond $\theta_{R'_{\rm max}}$ because the NFW neighbour has a finite $R_s \sim 0.3\,h^{-1}\,{\rm Mpc}$.

$\xi_{\rm NL}$-decay is not the binding constraint. One might worry that $\xi_{\rm NL}(r)$ support extends to tens of $h^{-1}\,{\rm Mpc}$ and therefore forces $\theta_{\rm max}$ large even for $\langle\Delta\Sigma^{\rm prj}\rangle$. It does not, for the same reason the `rnd` plateau drops out: at large θ the $\Delta\Sigma_{\rm mis}$-kernel is what zeros the integrand, not $\xi_{\rm NL}$. $\xi_{\rm NL}$ decay would bind $\theta_{\rm max}$ only for an observable where $\Delta\Sigma_{\rm mis}$ is replaced by an R'-independent kernel (e.g. the $\langle\Sigma^{\rm prj}\rangle$ `rnd` piece). Here, taking $\theta_{\rm max}$ out past $\theta_{R'_{\rm max}} \times$(few) is wasted work.

Regression check. An analogue of `validations/sigma_prj_theta_max.py` for the $\Delta\Sigma^{\rm prj}_{\rm cl+LSS}$ observable should show:

- moving $\theta_{\rm max} : R_{\rm max} = 30 \to 60\,h^{-1}\,{\rm Mpc}$ leaves $\Delta\Sigma^{\rm prj}_{\rm cl+LSS}(R')$ unchanged to $\ll 1\%$ on every $R' \le R'_{\rm max}$,
- reducing $\theta_{\rm max} : R_{\rm max} = 30 \to 3\,R'_{\rm max}$ ($\approx 9\,h^{-1}\,{\rm Mpc}$ for $R'_{\rm max} = 3$) also leaves $\Delta\Sigma^{\rm prj}_{\rm cl+LSS}$ unchanged to $\ll 1\%$,
- reducing further (e.g. $R_{\rm max} = R'_{\rm max}$) starts to clip the ring-peak tail at $R_{\rm mis} = R'_{\rm max}$ and produces a $\gtrsim 1\%$ bias at $R' \sim R'_{\rm max}$.

The $C = 3$ floor in (14) is set to leave a healthy margin before that last regime.

4 What changes in the code

In `sigma_prj.py`, the only functional change needed to compute $\langle\Delta\Sigma^{\rm prj}(R')\rangle$ is to replace the kernel lookup

`NFWMiscentered.sigma_grid(R, Rtheta, M, z)` $\longrightarrow$ `NFWMiscentered.delta_sigma_grid(R, Rtheta, M, z)` where `delta_sigma_grid` returns $\bar{\Sigma}_{\rm mis}(< R' \mid M, z, R_{\rm mis}) - \Sigma_{\rm mis}(R' \mid M, z, R_{\rm mis})$ at the same offset-NFW lookup resolution and with the same factor-of-2 convention as `sigma_grid`. The second change is to make the θ-upper limit adaptive to the requested R'-grid rather than fixed at `R_MAX_CMPCH`: set $R^{\Delta\Sigma}_{\rm max} = C \cdot R'_{\rm max}$ with $C = 3$ (eq. 14), plumbed into the θ-grid constructor in place of the hard-coded $30\, h^{-1}$ Mpc. Everything else — the θ-outer loop, the split-at-breakpoints θ-grid with per-R' nodes, the ring $\cup$ outer-fg $\cup$ outer-bg z-grid, $b_{\rm sel}$ evaluation — is preserved, and the `cl+LSS/rnd` decomposition becomes moot because the default return is `cl+LSS` only (eq. 13).

References

Cooray, A., & Sheth, R. 2002, Phys. Rep., 372, 1. Eqs. 86–87 of the halo-model review.

Costanzi, M., Wu, H.-Y., Esteves, J. H., Grandis, S., To, C., & Aguena, M., 2026, Phys. Rev. D (accepted), arXiv:2604.05833. $b_{\rm sel}$, $\langle \Sigma^{\rm prj} \rangle$ (Eq. 13).

Esteves, J. H. 2026, `richness_selection` numerical recipe, `docs/sigma_prj_refactor.md`. $\langle \Sigma^{\rm prj} \rangle$ θ-grid, $R_{\rm max}$ sensitivity, `rnd/cl+LSS` decomposition.

Sheldon, E. S., et al. 2009, ApJ, 703, 2217. Random-point subtraction convention for cluster lensing.

Zu, Y., Weinberg, D. H., Rozo, E., Sheldon, E. S., Tinker, J. L., & Becker, M. R. 2014, MNRAS, 439, 1628.

Melchior, P., et al. 2017, MNRAS, 469, 4899. DES Y1 cluster lensing; uses random-point subtraction to remove $\Sigma_{\rm rnd}$.